

NUMERAI



NUMERAPI

the official Python API client

Community Meetup Prague

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Who am I

Wolfgang Steitz

- Living in Frankfurt, Germany
- PhD (Genetic Algorithms)
- Data Science in the travel industry
- Programmatic advertising
- Numerai on Fridays (uuazed)
- Maintainer of numerapi



What is numerAPI?

- Light Python wrapper around numerai's GraphQL API
- Including some convenience functions
- Straight-forward interaction with numerai
- Aims to be robust and simple

- numerapi.readthedocs.io
- github.com/uuazed/numerapi

Total downloads

511.331

Total downloads - 30 days

16.775

Total downloads - 7 days

3.835

numerAPI is a community project



numerapi

Public



Pin



Unwatch

18



Fork

66



Star

165



- Open source project hosted on Github
- 30+ contributors
- Created 2017
- Considered as the official Python library by numerai

```
pip install numerapi
```

Fetching data

```
[2] from numerapi import NumerAPI
     napi = NumerAPI()

     # v4.1 is the latest version of the dataset
     [f for f in napi.list_datasets() if f.startswith("v4.1")]

['v4.1/features.json',
 'v4.1/live.parquet',
 'v4.1/live_example_preds.csv',
 'v4.1/live_example_preds.parquet',
 'v4.1/live_int8.parquet',
 'v4.1/meta_model.parquet',
 'v4.1/train.parquet',
 'v4.1/train_int8.parquet',
 'v4.1/validation.parquet',
 'v4.1/validation_example_preds.csv',
 'v4.1/validation_example_preds.parquet',
 'v4.1/validation_int8.parquet']
```

Fetching data & train a simple model

```
import pandas as pd
import lightgbm as lgb
import json

napi.download_dataset("v4.1/train.parquet")
napi.download_dataset("v4.1/features.json")

# We will be using the "small" feature set to save time and memory
# In practice, you will likely want to use all the features to maximize your model's performance
feature_metadata = json.load(open("v4.1/features.json"))
feature_cols = feature_metadata["feature_sets"]["small"]

# load the data
training_data = pd.read_parquet("v4.1/train.parquet", columns= ["era"] + feature_cols + ["target"])

# train a simple model
model = lgb.LGBMRegressor(n_estimators=20, learning_rate=0.01, max_depth=5, colsample_bytree=0.1)
model.fit(training_data.filter(like="feature"), training_data["target"])
```

Some actions require authentication

API keys to programmatically interact with Numerai services

Create API Key

```
napi = numerapi.NumerAPI("<public id>", "<token>")
```

```
export NUMERAI_PUBLIC_ID=123  
export NUMERAI_SECRET_KEY=ABC  
|
```


Upload predictions

```
# Load latest live features
napi.download_dataset("v4/live.parquet")
live_features = pd.read_parquet(f"v4/live.parquet", columns=feature_cols)

# Generate live predictions with our trained model
live_predictions = model.predict(live_features)

# Format submission
submission = pd.Series(live_predictions, index=live_features.index).to_frame("prediction")
submission.to_csv("preds.csv")

# uploading requires authentication!
napi = numerapi.NumerAPI("<public id>", "<token>")

model_id = napi.get_models()['uuazed']
napi.upload_predictions("preds.csv", model_id=model_id)
```

Diagnostics works the same way

```
napi.upload_diagnostics("preds.csv", model_id=model_id)
```

Fetching scores

```
napi.round_model_performances("uuazed")
```

```
napi.get_leaderboard()
```

Pandas DataFrames can be created easily from most responses.

```
pd.DataFrame(napi.round_model_performances("uuazed"))
```



NUMERAI SIGNALS

```
import numerapi

napi = numerapi.SignalsAPI("<public id>", "<token>")

model_id = napi.get_models()['uuazed']
napi.upload_predictions("preds.csv", model_id=model_id)

napi.round_model_performances("uuazed")
```

Basic stake management – skin in the game

```
] # set amount of NMR  
  napi.stake_set(10000, model_id=model_id)  
  
# release your full stake  
napi.stake_drain(model_id=model_id)
```

Automation

```
import time

while not napi.check_round_open():
    time.sleep(60 * 10)
...
```

- Not recommended
- Better to use webhooks or Numerai compute

Write your own GraphQL queries

- <https://api-tournament.numer.ai/>

GraphiQL  History Save

```
1 query {
2   pipelineStatus (date:"2021-09-16"){
3     dataReadyAt
4     resolvedAt
5     scoredAt
6     tournament
7     scoredAt
8   }
9 }
10 }
```

```
{
  "data": {
    "pipelineStatus": {
      "dataReadyAt":
        "2021-09-16T12:44:53Z",
      "resolvedAt": "2021-09-16T17:34:27Z",
      "scoredAt": "2021-09-16T16:32:40Z",
      "tournament": "classic"
    }
  }
}
```

QUERY VARIABLES

Documentation Explorer

 Search Schema...

A GraphQL schema provides a root type for each kind of operation.

ROOT TYPES

query: RootQueryType

mutation: RootMutationType

Write your own GraphQL queries

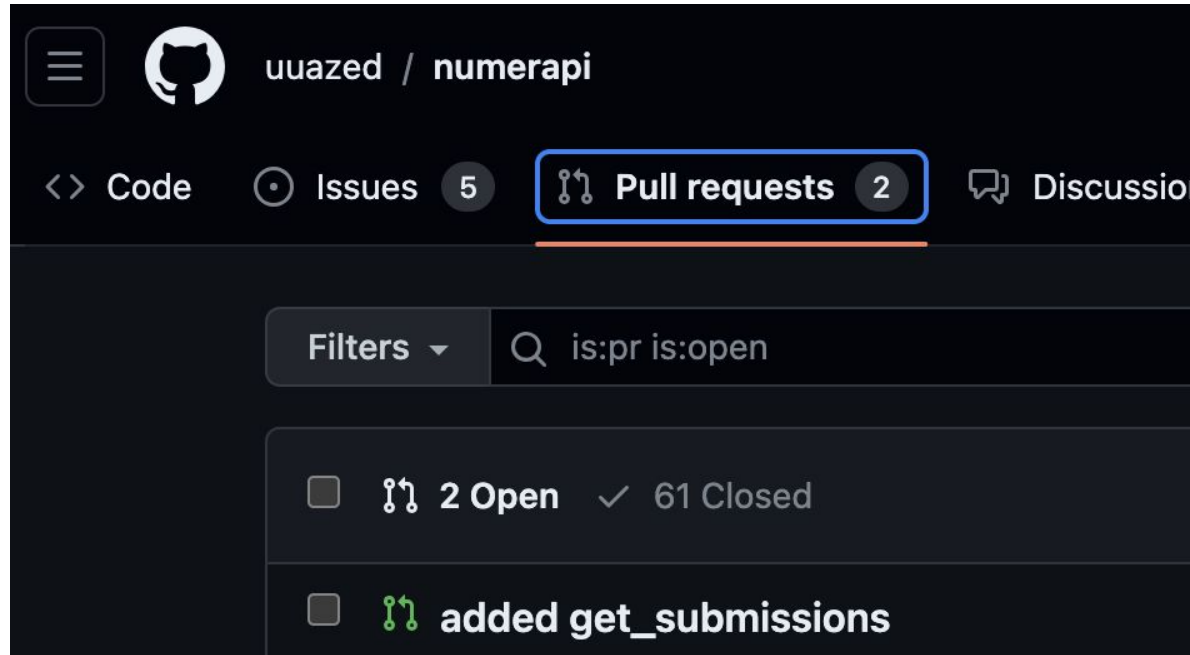
```
query = ""  
query {  
  pipelineStatus (date:"2021-09-16"){  
    dataReadyAt  
    resolvedAt  
    scoredAt  
    tournament  
    scoredAt  
  }  
}  
}""  
napi.raw_query(query, authorization=False)
```

```
{'data': {'pipelineStatus': {'dataReadyAt': '2021-09-16T12:44:53Z',  
  'resolvedAt': '2021-09-16T17:34:27Z',  
  'scoredAt': '2021-09-16T16:32:40Z',  
  'tournament': 'classic'}}}
```


Command line interface

```
(p3) → ~  
(p3) → ~ numerapi --help  
Usage: numerapi [OPTIONS] COMMAND [ARGS]...  
  
    Wrapper around the Numerai API  
  
Options:  
  --help  Show this message and exit.  
  
Commands:  
  account                Get all information about your account!  
  check-new-round        Check if a new round has started within the...  
  competitions            Retrieves information about all competitions  
  current-round          Get number of the current active round.  
  daily-model-performances  Fetch daily performance of a model.  
  dataset-url            Fetch url of the current dataset.  
  download-dataset        Download specified file for the given round  
  download-dataset-old    Download dataset for the current active round.  
  leaderboard            Get the leaderboard.  
  list-datasets          List of available data files  
  models                Get map of account models!  
  profile                Fetch the public profile of a user.  
  stake-decrease         Decrease your stake by `value` NMR.  
  stake-drain            Completely remove your stake.  
  stake-get              Get stake value of a user.  
  stake-increase         Increase your stake by `value` NMR.  
  submission-filenames   Get filenames of your submissions  
  submit                Upload predictions from file.  
  transactions           List all your deposits and withdrawals
```

Contributions are welcome



Thanks!

- Github: @uuazed
- Mail: uuazed@gmail.com
- Discord: uuazed
- Linkedin: wsteitz



NUMERA I